App Development 

Junior Coding for Good Badge — Girls Build Bots

Presented by your robotics mentors!

Start with a USER 💚

App developers never start with the code.
They start with **a person and their problem**.
This is the purest “for good” badge in the series.

Reverse-engineer an app 🔍

Look at an app you know. Ask the designer questions:

- **Who is the user?**
- **What problem does it solve?**
- **What's the very first thing you see — and why?**

Big buttons for important actions. The main thing front and center.

Apps are designed, not accidental.

Interview your user 🎤

Your team draws a **user card**. A mentor plays that person.

Ask them: - “Walk me through your day.” - “When does the problem happen?” - “What have you tried?”

Write the problem statement 

[User] needs a way to [goal] because [barrier].

Developers who skip this step **build apps nobody needs**.

Wireframe the app ✏

Paper prototype, 4–5 screens:

1. **Welcome** — name + what it does, *in the user's words*
2. **Main screen** — the one button that solves the problem (make it BIG)
3. **Payoff** — what happens after the tap
4. **Extra feature** — *only if the interview justifies it*

The developer’s question

As you design, we’ll keep asking:

“Which screen solves the problem statement?”

If a screen doesn’t serve the user, **cut it**. Real developers cut features constantly.

Click Test 👆

Your user comes back!

1. The user taps your paper buttons
2. Your team swaps screens by hand
3. **Does the user solve their problem in three taps or fewer?**

Then make one fix based on what confused them.

Robot moment 

Our team designs the **driver dashboard** the same way.

The driver is the user. The interface shows exactly what matters mid-match — **nothing more**.

Present your app 🎤

One sentence:

“***[App name]* helps *[user]* to *[solve problem]*.**”

What you accomplished ✅

- Identified a real app’s user and the problem it solves
- Interviewed a user and wrote a problem statement
- Designed a wireframe where every screen serves the user
- Ran a click test and improved from real feedback

Keep going at home! 

- Interview someone in your family about a problem in their day
- Wireframe the fix on paper — three taps or fewer
- Ask your troop leader about the next Coding for Good badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.